Robotics Lab PCs — Environment Setup Guide

Oct 28, 2026 · @Akram Fatayer

This guide installs one shared, read-only Python environment called `zuj-robotics` on every PC used for the Intelligent Mobile Robots (0135343) labs. Both sections and every student account then use the same environment, with nothing to install during lab time. Building the package takes about 30 minutes, once per semester; installing it takes about 3 minutes per PC.

Before you start

You need:

Item	Detail
Administrator account	On every lab PC (the install writes to `C:\envs` and the Public Desktop)
Lab PCs	Windows 10 (version 1803 or later) or Windows 11, 64-bit; at least 2 GB free on `C:`
One build PC	Any Windows 10/11 64-bit PC with internet access; it can be a lab PC
USB stick	At least 1 GB, or a network share every lab PC can reach
About 30 minutes	For Part A, once per semester
About 3 minutes per PC	For Parts B and C

Check these before the first PC:

- **Deep Freeze or Reboot Restore:** if the PCs use one, put them in thawed or maintenance mode first. Otherwise the install disappears at the next restart. Freeze them again after Part C.
- **Student accounts:** students should log in with standard (non-administrator) accounts. The environment is read-only only for standard accounts, so a shared admin account lets students break it.
- **Antivirus:** real-time scanning can slow the extraction step to several minutes. This is normal; don't cancel it.

- **Ready-made package:** the instructor normally gives you the `lab-pc` folder with `zuj-robotics.tar.gz` inside, ready to install and tested on 28 October 2026. Start with its START HERE.txt page, skip Part A and go straight to Part B.

Part A: Build the package (once per semester)

Part A builds the environment once, on one PC with internet, and packs it into a single file of about 480 MB. Lab PCs then need no internet and no Python or Anaconda of their own.

1. **Install Miniforge** on the build PC. Download `Miniforge3-Windows-x86_64.exe` from github.com/conda-forge/miniforge, run it and keep the default options. If the PC already has Anaconda or Miniconda, skip this step and use the Anaconda Prompt instead.
2. **Get the course files.** Open github.com/akramfatayer/zuj-robotics, click **Code → Download ZIP**, and extract it to `C:\build`. You should now have the folder `C:\build\zuj-robotics-main`.
3. **Open Miniforge Prompt** from the Start menu, and go to that folder:

   ```
   cd /d C:\build\zuj-robotics-main
   ```

4. **Create the environment.** This downloads about 500 MB and takes 5 to 10 minutes:

   ```
   conda env create -f environment.yml
   ```

   If it says the environment already exists (from an earlier semester), remove it with `conda env remove -n zuj-robotics -y` and run the command again.
5. **Test it.** A PyBullet window should open for 3 seconds with a robot in it, and every line should say `[OK]`:

   ```
   conda activate zuj-robotics
   python check_env.py --gui
   ```

   One line saying `pybullet build time: ...` is normal.
6. **Install the packing tool** into the base environment:

   ```
   conda deactivate
   conda install -n base -c conda-forge conda-pack -y
   ```

7. **Pack the environment** into the `lab-pc` folder (about 1 minute):

   ```
   conda pack -n zuj-robotics -o lab-pc\zuj-robotics.tar.gz
   copy check_env.py lab-pc\
   ```

8. **Copy the whole `lab-pc` folder** to the USB stick or network share. It must contain these files:

File	What it is
`zuj-robotics.tar.gz`	The packed environment, about 480 MB
`deploy_lab_pc.bat`	The installer you run on each PC
`Robotics-Lab.bat`	The desktop shortcut students use; the installer copies it
`check_env.py`	The checker; the installer copies it
`verify_lab_pc.bat`	The student-account check for Part C; the installer copies it to C:\envs
`START HERE.txt`	A one-page summary of Parts B and C, with the package's size and SHA256 checksum

Part B: Install on each lab PC (about 3 minutes)

Repeat these steps on every lab PC. If IT uses a master disk image, do Parts B and C once on the master PC and capture the image instead.

1. **Log in** with an administrator account.
2. **Copy the `lab-pc` folder** from the USB stick to `C:\Temp\lab-pc`. Running the installer straight from a network drive often fails, because drives mapped as a normal user are not visible to programs running as administrator.
3. **Run the installer.** In `C:\Temp\lab-pc`, right-click `deploy_lab_pc.bat`, choose **Run as administrator**, and click **Yes** in the Windows prompt.
4. **Wait** while the window shows five steps. Step 2, the extraction, is the slow one: 1 to 3 minutes.
5. **Read the result.** The window should end with `All checks passed` and `This PC is ready.` Press any key to close it. If it ends with `DEPLOYMENT FINISHED WITH ERRORS`, note the `[FAIL]` lines and see Troubleshooting.
6. **Delete** `C:\Temp\lab-pc` to free 480 MB.

The installer changes only these things on the PC. It doesn't touch PATH, the registry, or any existing Python or Anaconda install.

Location	What is there
`C:\envs\zuj-robotics`	The environment: Python 3.11, PyBullet, NumPy, Matplotlib, OpenCV
`C:\envs\check_env.py`, `C:\envs\verify_lab_pc.bat`	The checker that runs each time a student opens Robotics Lab, and the student-account check for Part C
`C:\Users\Public\Desktop\Robotics Lab.bat`	The **Robotics Lab** shortcut, visible to every account
Permissions on `C:\envs`	Administrators and SYSTEM: full control. Users (students): read and run only

Running the installer again is safe. It removes the old copy and installs a fresh one, which is also how you repair a broken PC.

Part C: Verify each PC as a student

The installer's own test runs as administrator, so check what a student actually sees. Log out, log in with a **student (standard) account**, and tick each item:

- ☐ **Shortcut: Robotics Lab** is on the desktop. Double-click it: a window titled `Robotics Lab (zuj-robotics)` opens, every line says `[OK]`, and it ends with the reminder to save work off the PC.
- ☐ **Student check:** double-click `C:\envs\verify_lab_pc.bat`. It checks that the shortcut is there, that the account is a standard one, that students cannot change the environment, and every package. A robot window opens for 3 seconds, then a plot window: close the plot. It must end with This PC passed. Watch for the robot window on at least one PC of each hardware model, because it depends on the graphics driver.
- ☐ **Done:** log out, and freeze the PC again if it uses Deep Freeze or Reboot Restore.

A student running `pip install` on a lab PC may see `Defaulting to user installation`. That is harmless: the shortcut tells Python to ignore packages installed that way, so the course environment stays the same for everyone.

Deployment tracker

One row per PC. Set the status to **Verified** only after every Part C check passes. Add rows for the remaining PCs.

PC	Room	Status	3D window OK	Done by	Notes
PC 01		Not started			
PC 02		Not started			
PC 03		Not started			
PC 04		Not started			
PC 05		Not started			
PC 06		Not started			

Troubleshooting

Find the message you see in the first column. Re-running the installer (Part B) fixes most problems on a single PC.

What you see	Likely cause	Fix
`CondaHTTPError` or a timeout during Part A	The university proxy or firewall blocks conda	Build on another network, or ask IT for the proxy address and add it to conda's settings
"Windows protected your PC" when running a `.bat`	Files downloaded from the internet are marked as untrusted	Click **More info → Run anyway**, or right-click the file → **Properties** → tick **Unblock**
`Please right-click this file and choose "Run as administrator"`	The installer was double-clicked	Right-click it → **Run as administrator**
`zuj-robotics.tar.gz was not found next to this script`	The folder was copied incompletely, or is still zipped	Copy the whole `lab-pc` folder again; extract it first if it is a ZIP
`Could not set permissions on C:\envs`	No administrator rights, or a security policy blocks the change	Use a local administrator account; if it still fails, ask IT

What you see	Likely cause	Fix
`'tar' is not recognized`	Windows 10 older than version 1803	Update Windows, or extract the package with 7-Zip into `C:\envs\zuj-robotics`, then run `C:\envs\zuj-robotics\Scripts\conda-unpack.exe`
`Extraction failed`	Less than 2 GB free on `C:`, or a damaged copy	Free space; copy `zuj-robotics.tar.gz` again and check its size matches the original
`[FAIL]` lines, or `DLL load failed`	Antivirus quarantined a file during extraction	Restore the file, exclude `C:\envs` from scanning, run the installer again
Students see `The robotics environment is not installed on this PC`	Deep Freeze removed it, or the PC was missed	Thaw the PC, run the installer again, freeze it
No **Robotics Lab** icon for students	Desktop redirected to OneDrive, or hidden by policy	Copy `C:\Users\Public\Desktop\Robotics Lab.bat` to the Start menu, or tell students to open it from there
The 3D window doesn't open, or closes at once	Old graphics driver, or you are connected by Remote Desktop	Test sitting at the PC, not over Remote Desktop; update the graphics driver. Students can still finish labs without the 3D window
`verify_lab_pc.bat` says students can change the environment	The student account is an administrator	Give students standard accounts, or accept that they can change the environment and re-run the installer when needed
`[FAIL] NumPy matrix maths (np.linalg) crashed`	Python was started without the Robotics Lab shortcut, or antivirus removed one of NumPy's maths files	Run it from the Robotics Lab window. If it still fails there, exclude C:\envs from scanning and run the installer again

Updating, repairing and removing

Students keep their work on GitHub or OneDrive, never in `C:\envs`, so none of these steps affect student files.

- **Repair one PC:** run the installer again (Part B). It replaces the environment completely.
- **Update all PCs** (when the instructor announces a new version): download the latest course files, remove the old build with `conda env remove -n zuj-robotics -y`, repeat Part A, then Parts B and C on every PC.
- **Remove at the end of the semester:** open Command Prompt as administrator and run:

```
rmdir /s /q C:\envs\zuj-robotics
del C:\envs\check_env.py C:\envs\verify_lab_pc.bat
del "C:\Users\Public\Desktop\Robotics Lab.bat"
```

Optional extras

Neither is required; the labs run from the **Robotics Lab** window alone.

- **Git for Windows** (git-scm.com, default options): lets students clone the course and push their own work at the end of each session, which is the safest way to keep work on shared PCs.
- **Visual Studio Code** (code.visualstudio.com): fine for **editing** the lab files, but students should **run** them from the **Robotics Lab** window. Pointed straight at `C:\envs\zuj-robotics\python.exe`, without the shortcut, Python starts but NumPy's matrix functions (`np.linalg`, used by every Kalman filter lab) crash on load (tested October 2026).

Questions about this guide: Dr. Akram Fatayer, akram.fatayer@hotmail.com.